



According to some, the inspiration for cellophane came after its inventor, Jacques Brandenberger, saw wine spilled on a tablecloth. But the film proved more useful in waterproof packaging than for stain protection.

SWISS CHEMIST JACQUES E. BRANDENBERGER (1872–1954)

invented a way of producing sheets of thin waterproof film made from wood cellulose. The **film** came to be called **cellophane** (for cellulose and *diaphane*, French for transparent). Brandenberger's original idea was to spray liquefied cellulose onto fabric as a stain-repellent, but he found that he could pull away a dry film that was far more useful.

New Zealand-born physicist Ernest Rutherford's **half-life explanation of radioactivity** still commanded attention. He and German physicist Hans Geiger (1882–1945) had devised **a way of**

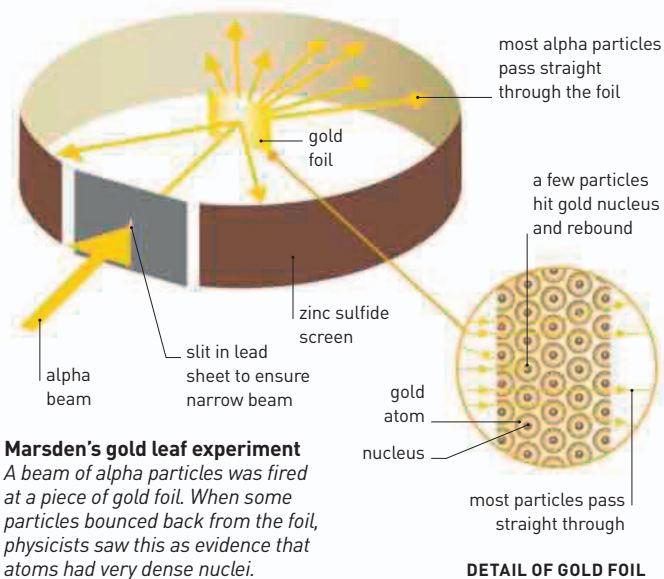
“IT WAS QUITE THE **MOST INCREDIBLE EVENT THAT HAS EVER HAPPENED TO ME IN MY LIFE.**”

Ernest Rutherford, New Zealand-born physicist, from his lecture *The Development of the Theory of Atomic Structure*, 1936

measuring radioactivity by scintillation—counting the flashes of light when the rays struck a zinc sulfide screen. They carried out experiments firing radiation through barriers, enlisting the help of a student called Ernest Marsden (1889–1970). Geiger and Marsden studied the effects of metallic

foils on a stream of alpha particles. Marsden's **work using gold foil threw up unexpected results**. According to the atomic structure theory of the time (the plum pudding model, see 1904) alpha particles should have passed straight through the gold, but instead a few bounced back. Rutherford later said it was, “as though you fired a bullet at tissue paper and it bounced back.” His analysis of Marsden's results suggested that the deflected particles were striking **a very dense nucleus at the core of each atom**.

In August, five years after their historic maiden flights, the **Wright brothers were in the air again**, but this time they had an audience. In an atmosphere of scepticism, Wilbur had traveled to France to show off their manned, powered aircraft. Over several days he demonstrated his mastery of flying and the assembled crowds grew daily: the Wright brothers became **aviation celebrities**.



A remarkable repository of some of the oldest animal fossils, Canada's Burgess Shale is a record of Cambrian ocean life half a billion years ago.

THE GERMAN DRUG COMPANY

Bayer—pioneer of aspirin—was granted a **patent on a sulfur-based drug**. The drug was a derivative of sulfonamides, a class of bacterial agents that would rise to prominence in 1932 and dominate the preantibiotic period. Sulfonamides marked a significant step in **chemotherapy**—the scientific development of drugs that cured disease using sound pharmaceutical principles—but Bayer was unaware of their importance. At the same time, German physician Paul Ehrlich and Japanese biologist **Sahachiro Hata** (1873–1978) were working on their own **custom-made drug** to treat the sexually transmitted bacterial disease **syphilis**.

Søren Sørensen (1879–1963), a Danish chemist, was studying protein—a substance that is sensitive to the effects of acids and alkali. Sørensen worked on a way of quantifying acidity and alkalinity and, as a result, **devised**

the pH scale. This scale rated substances according to whether they were acid (1–6), alkaline (8–14), or neutral (7).

The previous year, **Louis Blériot** (1872–1936), a French engineer, had witnessed the public manned flights of Wilbur Wright. Inspired into action, he set his sights on the English Channel. French inventor Jean Piere Blanchard had crossed it in a balloon (see 1785) but in July, Blériot became the **first person to cross the English Channel in a manned, powered plane**. The *Blériot XI* left Calais at sunrise on July 25 and landed in Dover 36 minutes later. As well as receiving international acclaim, Blériot won a £1,000 prize from London's *Daily Mail* newspaper.

In America, paleontologist **Charles Walcott** (1850–1927) made an **important discovery**.

Measuring pH value

Indicator paper contains a chemical that reacts with acid or alkali to produce a colour to match a scale according to the subject's “strength.”

